

Scalars & Vectors

①

- A scalar quantity is one that can be described using its magnitude. eg: time, temperature, mass.
- A vector is an entity/quantity that requires both magnitude & direction to describe it. eg: displacement, Velocity, force.
- Graphically represented as



↳ length gives the magnitude
↳ arrow indicates the direction

- For a vector to be zero, every component must be zero.
- For 2 vectors to be equal, both magnitude & direction must be same.
- Algebraically represented as

$$\vec{A} = \vec{A}_x + \vec{A}_y = A_x \hat{i} + A_y \hat{j}$$

magnitude $|\vec{A}| = \sqrt{A_x^2 + A_y^2}$

$\hat{i}$ - unit vector in x direction

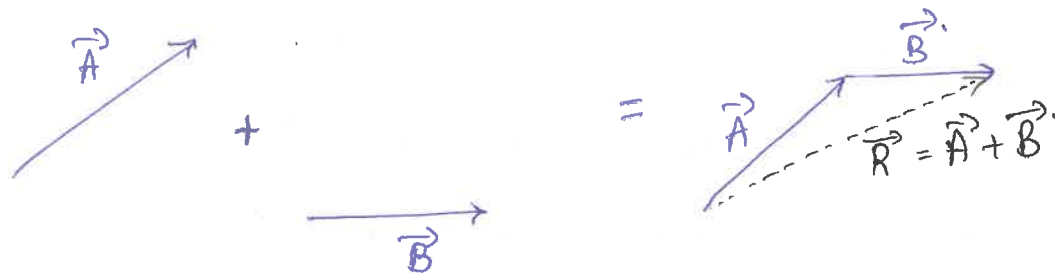
direction is given as

$$\theta = \tan^{-1} \left(\frac{A_y}{A_x} \right)$$

$\hat{j}$ - unit vector in y direction.

- Unit vector is a vector whose magnitude is '1'
- Unit vector of any vector $\vec{A}$ is given as $\frac{\vec{A}}{|\vec{A}|}$

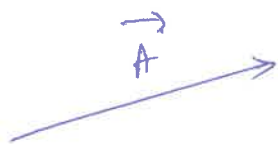
Addition of Vectors



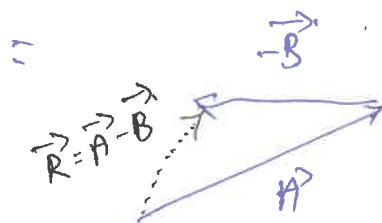
$$\vec{A} + \vec{B} = \vec{B} + \vec{A}$$

If we change the direction of vector opposite to its original direction, the sign of the vector changes

↳



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Subtraction of Vectors.

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Algebraically $\vec{A} \pm \vec{B} \left\{ \begin{array}{l} A_x \pm B_x \\ A_y \pm B_y \end{array} \right\}$ Component wise addition/subtraction.

$$\rightarrow \vec{A} + \vec{B} = \vec{B} + \vec{A}$$

vector addition is commutative.

$$\vec{A} - \vec{B} \neq \vec{B} - \vec{A}$$

$$\rightarrow (\vec{A} + \vec{B}) + \vec{C} = \vec{A} + (\vec{B} + \vec{C})$$

$$\rightarrow \underbrace{K}_{\text{scalar}} (\vec{A} + \vec{B}) = K\vec{A} + K\vec{B}$$

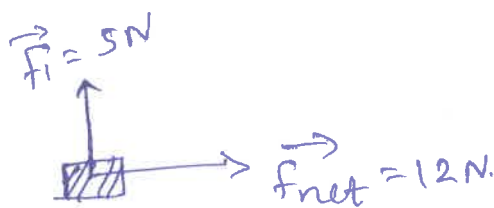
$$\rightarrow \vec{A} + 0 = \vec{A}$$

$$\rightarrow \vec{A} + (-\vec{A}) = 0$$

eg:-1: Given:  $\rightarrow F_{\text{net}} = 12 \text{ N}$ in east direction.

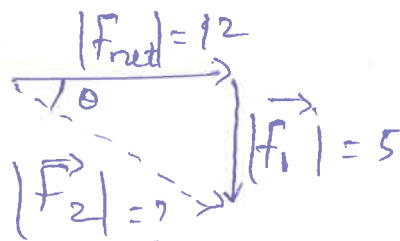
$\vec{F}_1$ & $\vec{F}_2$ act such that the resultant force is as above. $\vec{F} = 5 \text{ N}$ in north. What is $\vec{F}_2$?

Sol



$$\vec{F}_1 + \vec{F}_2 = \vec{F}_{\text{net}}$$

$$\vec{F}_2 = \vec{F}_{\text{net}} - \vec{F}_1$$



$$|\vec{F}_2| = \sqrt{12^2 + 5^2} = \underline{\underline{13 \text{ N}}}$$

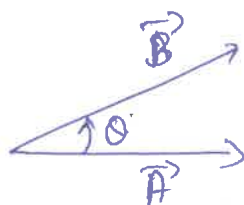
$\theta = \tan^{-1} \left(\frac{5}{12} \right)$ in South east direction.

Multiplication of Vectors

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I. Dot Product

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$$



$$\text{If } \vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$\& \vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|} \quad \left\{ \begin{array}{l} \text{angle between 2 vectors.} \end{array} \right.$$

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A} \quad \left\{ \begin{array}{l} \text{Commutative.} \end{array} \right.$$

$$\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C} \quad \left\{ \begin{array}{l} \text{distributive} \end{array} \right.$$

$$k (\vec{A} \cdot \vec{B}) = (k\vec{A}) \cdot \vec{B} = \vec{A} \cdot (k\vec{B}) \quad \left\{ \begin{array}{l} \text{associative} \\ \text{Scalar.} \end{array} \right.$$

eg: $\vec{A} = 2\hat{i} + \hat{j}$ $\vec{B} = \hat{i} + 3\hat{j}$ $\rightarrow$ angle between $\vec{A}$ & $\vec{B}$

$$\vec{A} \cdot \vec{B} = 2 \cdot 1 + 1 \cdot 3 = 2 + 3 = 5$$

$$|\vec{A}| = \sqrt{2^2 + 1^2} = \sqrt{5} \quad |\vec{B}| = \sqrt{1^2 + 3^2} = \sqrt{10}$$

$$\therefore \cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|} = \frac{5}{\sqrt{50}} = \frac{1}{\sqrt{2}} \Rightarrow \underline{\underline{\theta = 45^\circ}}$$

eg. Two vectors $\vec{A}$ & $\vec{B}$ are such that $\vec{A}$ makes $(\alpha_1, \alpha_2, \alpha_3)$ angles with x, y & z axis & $\vec{B}$ makes $(\beta_1, \beta_2, \beta_3)$ angles with x, y & z axis. Find the angle " θ " between $\vec{A}$ & $\vec{B}$ vectors in terms of $\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2$ & β_3 . (5)

Sol

$$\left. \begin{aligned} \vec{A}_x &= |A| \cos \alpha_1 \\ \vec{A}_y &= |A| \cos \alpha_2 \\ \vec{A}_z &= |A| \cos \alpha_3 \end{aligned} \right\} \text{Components of } \vec{A} \text{ vector in } x, y \text{ \& } z \text{ directions}$$

Similarly

$$\begin{aligned} \vec{B}_x &= |B| \cos \beta_1 \\ \vec{B}_y &= |B| \cos \beta_2 \\ \vec{B}_z &= |B| \cos \beta_3 \end{aligned}$$

$$\therefore \vec{A} \cdot \vec{B} = |A| \cdot |B| \cos \theta$$

$$\vec{A} \cdot \vec{B} = \vec{A}_x \cdot \vec{B}_x + \vec{A}_y \cdot \vec{B}_y + \vec{A}_z \cdot \vec{B}_z$$

$$\vec{A} \cdot \vec{B} = |A| \cdot \cos \alpha_1 |B| \cos \beta_1 + |A| \cos \alpha_2 |B| \cos \beta_2 + |A| \cos \alpha_3 |B| \cos \beta_3$$

$$\cancel{|A|} \cdot \cancel{|B|} \cos \theta = \cancel{|A|} \cdot \cancel{|B|} [\cos \alpha_1 \cos \beta_1 + \cos \alpha_2 \cos \beta_2 + \cos \alpha_3 \cos \beta_3]$$

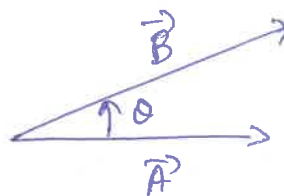
$$\therefore \cos \theta = \cos \alpha_1 \cos \beta_1 + \cos \alpha_2 \cos \beta_2 + \cos \alpha_3 \cos \beta_3$$

$$\theta = \cos^{-1} \{ \cos \alpha_1 \cos \beta_1 + \cos \alpha_2 \cos \beta_2 + \cos \alpha_3 \cos \beta_3 \}$$

II Cross Product

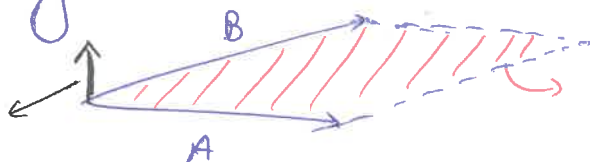
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$$\vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \sin \theta$$



Geometrically

direction of $\vec{A} \times \vec{B}$ is $\perp$ to both $\vec{A}$ & $\vec{B}$ vectors



Area of this parallelogram is $|\vec{A} \times \vec{B}|$ (magnitude)

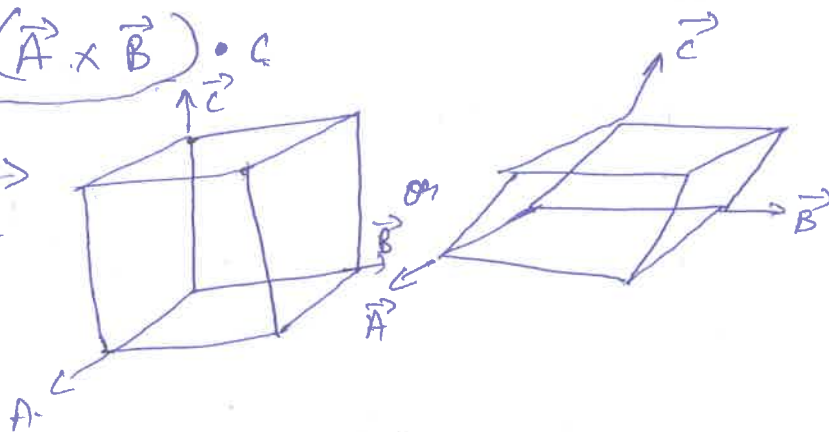
$$\rightarrow \vec{A} \times \vec{B} = -(\vec{B} \times \vec{A}) \text{ not commutative}$$

$$\rightarrow \vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C} \quad \text{distributive}$$

$$\rightarrow \vec{A} \times \vec{0} = \vec{0} \times \vec{A} = \vec{0}$$

$$\rightarrow \vec{A} \cdot (\vec{B} \times \vec{C}) = (\vec{A} \times \vec{B}) \cdot \vec{C}$$

Volume of these parallelepiped is given by these products



$$\rightarrow \vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C}) \vec{B} - (\vec{A} \cdot \vec{B}) \vec{C}$$

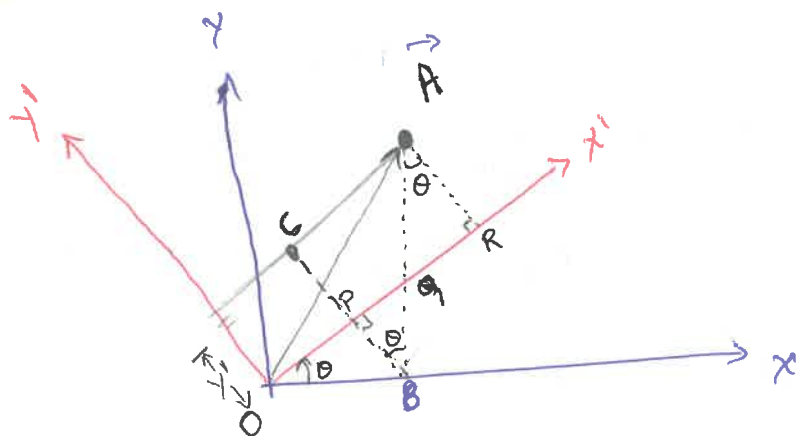
$$\rightarrow \vec{A} \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\vec{A} \times \vec{B}) = 0$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = \hat{i} (A_y B_z - A_z B_y) + \hat{j} (A_z B_x - A_x B_z) + \hat{k} (A_x B_y - A_y B_x)$$

In Cartesian Coordinates

$$\left. \begin{aligned} \hat{i} \cdot \hat{i} &= \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1 \\ \hat{i} \cdot \hat{j} &= \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0 \\ \hat{i} \times \hat{i} &= \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0 \\ \hat{i} \times \hat{j} &= \hat{k} ; \hat{j} \times \hat{k} = \hat{i} ; \hat{k} \times \hat{i} = \hat{j} \end{aligned} \right\}$$

Change of Vector Components under rotation.



Vector $\vec{A}$ in xy coordinates is given as

$$\vec{A} = x\hat{i} + y\hat{j} \text{ or } x + y \text{ where } \begin{matrix} x = x\hat{i} \\ y = y\hat{j} \end{matrix}$$

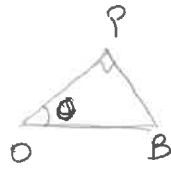
→ If Another axis exist about the same origin but is at an angle θ w.r.t to xy coordinates:
Then $\vec{A}$ can be written in the new coordinate system as

$$\vec{A} = x'\hat{i}' + y'\hat{j}'$$

$$X' = OR$$

$$= OP + PR$$

In $\triangle OBP$



$$OP = OB \cos \theta$$

& $OB = X$ (in unrotated frame)

$$\therefore OP = X \cos \theta$$

Now PR can be written as $PQ + QR$

$$\text{In } \triangle PBQ \quad PQ = PB \sin \theta$$

$$\& \text{ in } \triangle ARQ \quad QR = AQ \sin \theta$$

$$\therefore PR = PQ + QR = BQ \sin \theta + AQ \sin \theta = (BQ + AQ) \sin \theta$$

$$\text{but } BQ + AQ = Y$$

$$\therefore PR = Y \sin \theta$$

$$\therefore X' = OP + PR = X \cos \theta + Y \sin \theta$$

$$\therefore \text{In New Coordinates } \underline{X' = X \cos \theta + Y \sin \theta}$$

$$\text{In } \triangle ABC \quad AB \cos \theta = BC$$

$$Y \cos \theta = BP + PC$$

$$\text{Here } PC = AR = Y' \quad \& \quad \text{in } \triangle OPB \quad PB = OB \sin \theta = X \sin \theta$$

$$\therefore Y \cos \theta = X \sin \theta + Y'$$

$$\therefore \underline{Y' = -X \sin \theta + Y \cos \theta}$$

$$\therefore X' = X \cos \theta + Y \sin \theta$$

$$Y' = -X \sin \theta + Y \cos \theta.$$

→ Two vectors $\vec{A}$ & $\vec{B}$ in XY coordinate system are given by $\vec{A} = \vec{A}_x + \vec{A}_y$
& $\vec{B} = \vec{B}_x + \vec{B}_y$

→ The same vectors in $X'Y'$ coordinates obtained by rotating XY by angle θ are given as

$$\left. \begin{aligned} \vec{A}_x' &= \vec{A}_x \cos \theta + \vec{A}_y \sin \theta \\ \vec{A}_y' &= -\vec{A}_x \sin \theta + \vec{A}_y \cos \theta \end{aligned} \right\} \vec{A}' = \vec{A}_x' + \vec{A}_y'$$

$$\text{Similarly } \left. \begin{aligned} \vec{B}_x' &= \vec{B}_x \cos \theta + \vec{B}_y \sin \theta \\ \vec{B}_y' &= -\vec{B}_x \sin \theta + \vec{B}_y \cos \theta \end{aligned} \right\} \vec{B}' = \vec{B}_x' + \vec{B}_y'$$

In the rotated frame.

$$\vec{A}' \cdot \vec{B}' = \vec{A}_x' \cdot \vec{B}_x' + \vec{A}_y' \cdot \vec{B}_y'$$

$$= (\vec{A}_x \cos \theta + \vec{A}_y \sin \theta) (\vec{B}_x \cos \theta + \vec{B}_y \sin \theta)$$

$$+ (-\vec{A}_x \sin \theta + \vec{A}_y \cos \theta) (-\vec{B}_x \sin \theta + \vec{B}_y \cos \theta)$$

$$= A_x B_x \cos^2 \theta + A_y B_y \sin^2 \theta + A_x B_y \cancel{\cos \theta} \sin \theta + A_y B_x \cancel{\cos \theta} \sin \theta$$

$$+ A_x B_x \cos^2 \theta - A_x B_y \cancel{\cos \theta} \sin \theta - A_y B_x \cancel{\cos \theta} \sin \theta$$

$$+ A_y B_y \sin^2 \theta$$

$$\vec{A}' \cdot \vec{B}' = (\vec{A}_x \vec{B}_x + \vec{A}_y \vec{B}_y) (\cos^2 \theta + \sin^2 \theta)$$

$$\vec{A}' \cdot \vec{B}' = \vec{A}_x \vec{B}_x + \vec{A}_y \vec{B}_y$$

$$\boxed{\vec{A}' \cdot \vec{B}' = \vec{A} \cdot \vec{B}}$$

This is dot product
of A & B in x, y coordinate
system

This shows dot product of two vectors remains unchanged under rotation.

While.

$$\vec{A}' \times \vec{B}' = \underbrace{\vec{A}' \times \vec{B}'}_{x' \text{ component}} \bigg|_{x'} + \underbrace{\vec{A}' \times \vec{B}'}_{y' \text{ component}} \bigg|_{y'}$$

$$\vec{A}' \times \vec{B}' \bigg|_{x'} = (\vec{A} \times \vec{B})_x \cos \theta + (\vec{A} \times \vec{B})_y \sin \theta$$

$$\vec{A}' \times \vec{B}' \bigg|_{y'} = -(\vec{A} \times \vec{B})_x \sin \theta + (\vec{A} \times \vec{B})_y \cos \theta$$

$$\vec{A}' \times \vec{B}' = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} (\vec{A} \times \vec{B})$$

→ This indicates vector quantities change & resolve similar to coordinates under rotation.